



Virtual Rock Kit

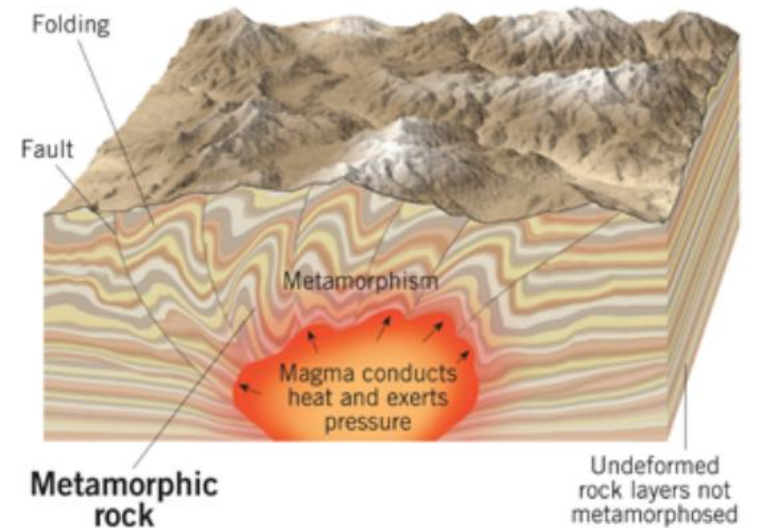
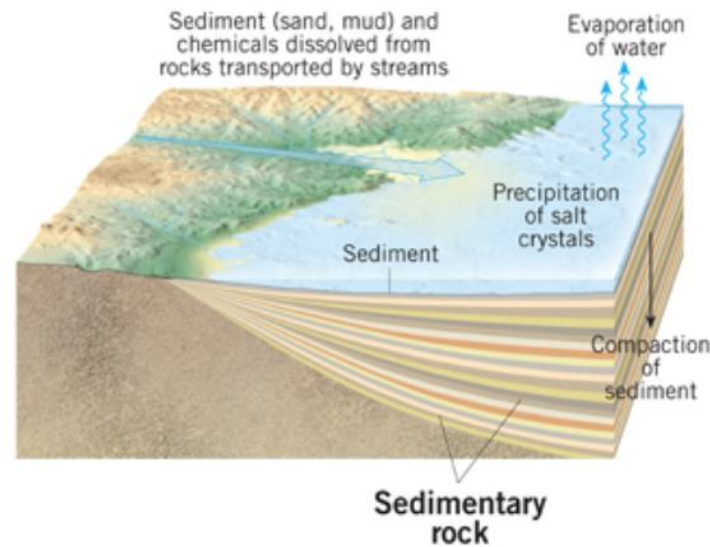
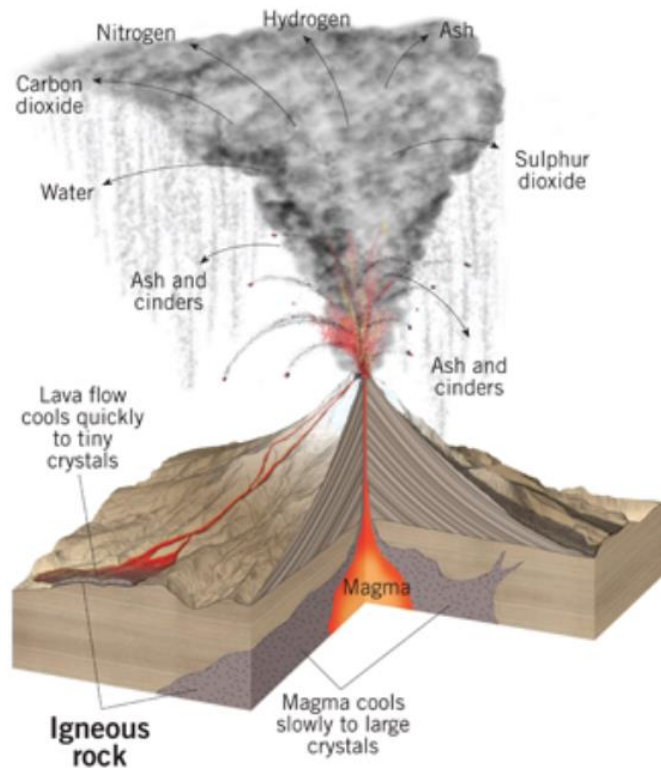


Igneous Rocks



Overview

- Geologists divide rocks into three major groups: igneous, sedimentary, and metamorphic





Overview

- Despite the difference in how these three rock types form, they are identified the same way – by characterizing their:
 - Texture – the overall appearance of a rock based on the size, shape, and arrangement of its component(s)
 - Composition – the components (minerals and/or clasts) that make up the rock



Objectives

- Characterize the texture and composition of igneous, sedimentary, and metamorphic rocks in your virtual rock kit;
- Identify their proper rock name using the provided identification tables.
 - The rock names highlighted and in bold uppercase font on the provided identification tables are provided in this virtual rock kit:

EXAMPLE

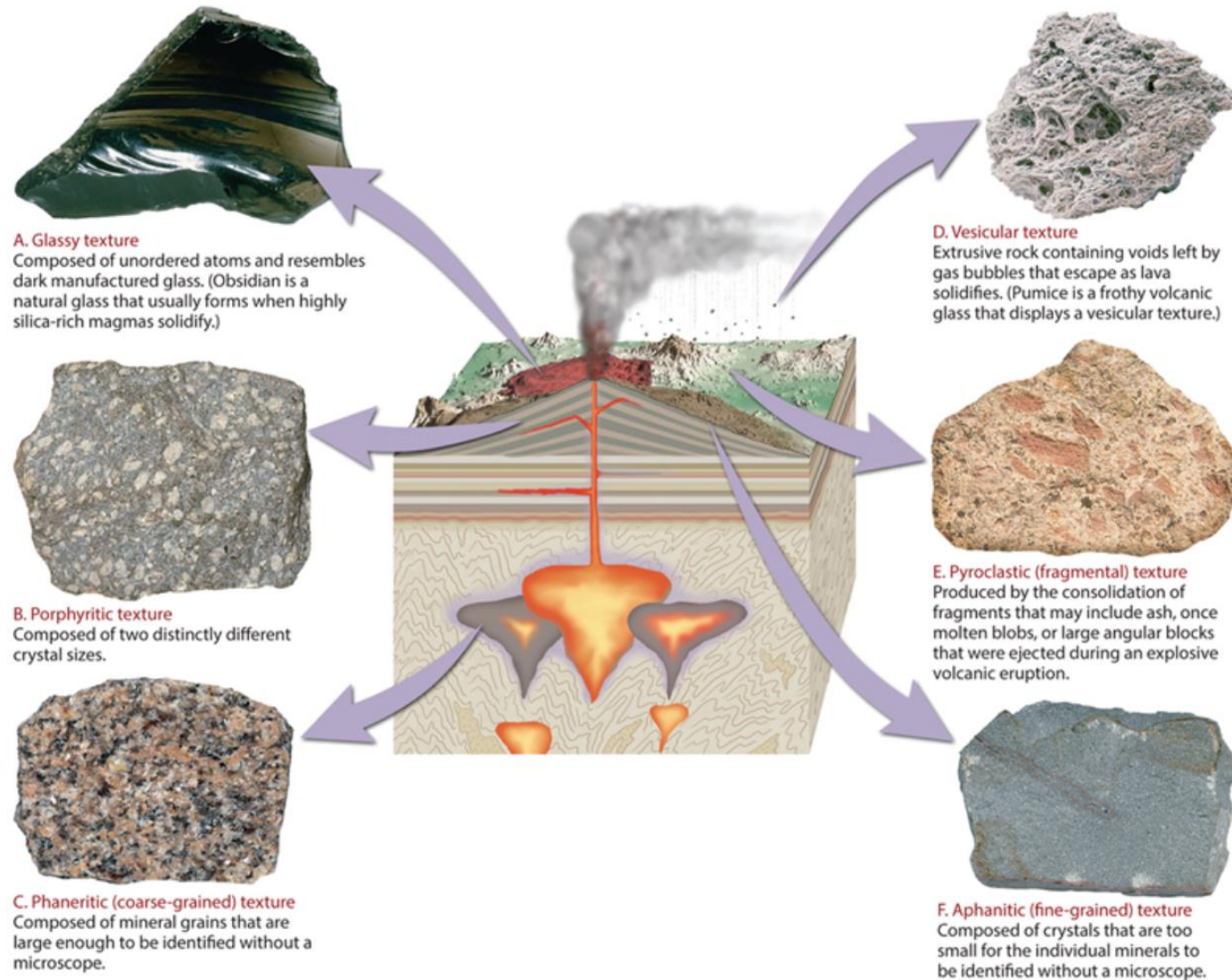


Igneous Rock Identification

- Texture – The texture of an igneous rock tells us its rate of cooling
 - Intrusive rocks cool *slowly* underground
 - Slow cooling promotes the growth of larger crystals (visible to the naked eye)
 - Intrusive texture terms: phaneritic (primary), porphyritic (secondary)
 - Extrusive rocks cool *quickly* above ground
 - Quick cooling results in fewer, smaller crystals (typically not visible to the naked eye)
 - Extrusive texture terms: Aphanitic, pyroclastic, glassy, vesicular



Igneous Rock Identification





Igneous Rock Identification

- Composition – The color of an igneous rock tells us its composition
 - A combination of 8 minerals form most igneous rocks:
 - Olivine, pyroxene, amphibole, biotite (dark-colored minerals) (*mafic* minerals)
 - Plagioclase feldspar, potassium feldspar, quartz, muscovite (light-colored minerals) (*felsic* minerals)
 - Igneous rocks are grouped into four compositional categories based on their mafic mineral percentages:
 - Ultramafic (85% - 100% mafic minerals)
 - Mafic (45% - 85% mafic minerals)
 - Intermediate (15% - 45% mafic minerals)
 - Felsic (0% - 15% mafic minerals)



Igneous Rock Identification

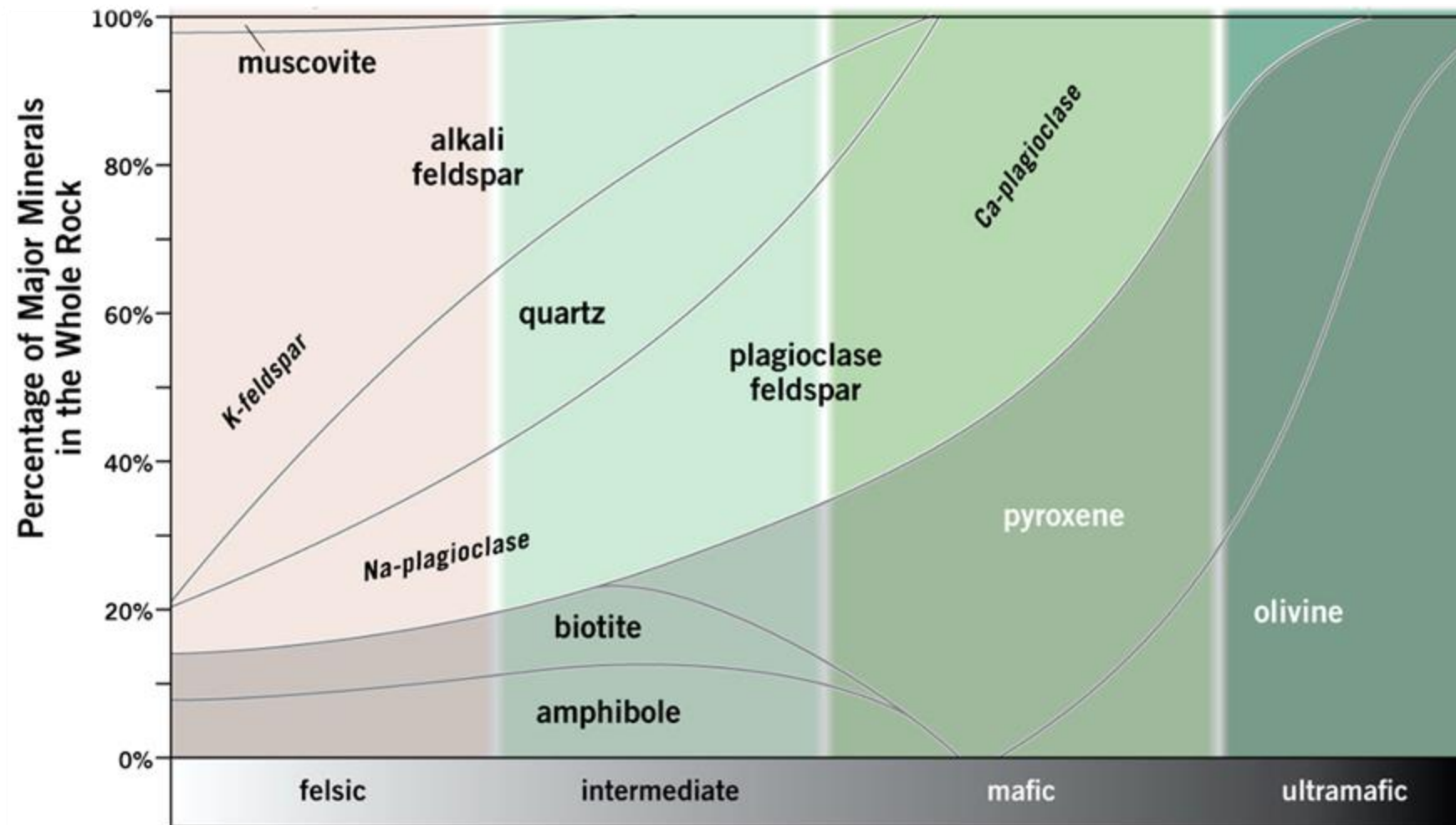


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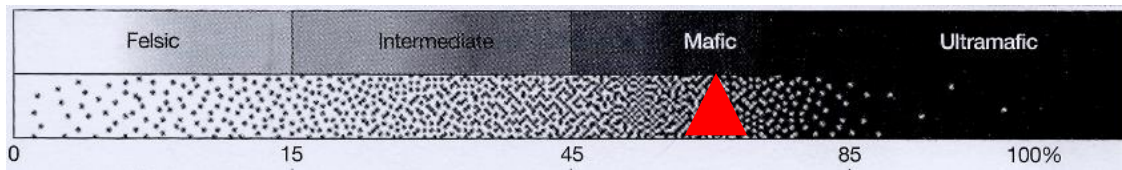
Igneous Rock Identification

Texture		Composition (Mafic Color Index)			
		Felsic	Intermediate	Mafic	Ultramafic
Intrusive Origin	Pegmatitic (<i>very coarse-grained</i>)	Pegmatitic Granite	Pegmatitic Diorite	Pegmatitic Gabbro	Pegmatitic Peridotite
	Phaneritic (<i>coarse-grained</i>)	GRANITE	Diorite	GABBRO	Peridotite
	Porphyritic (<i>phaneritic groundmass</i>)	Porphyritic Granite	Porphyritic Diorite	Porphyritic Gabbro	Porphyritic Peridotite
Extrusive Origin	Porphyritic (<i>aphanitic groundmass</i>)	Porphyritic Rhyolite	Porphyritic Andesite	Porphyritic Basalt	Komatite (<i>rare</i>)
	Aphanitic (<i>fine-grained</i>)	RHYOLITE	Andesite	BASALT	
	Glassy	OBSIDIAN			
	Vesicular	Pumice		SCORIA (many vesicles)	
				Vesicular Basalt (few vesicles)	
	Pyroclastic (<i>rock fragments</i>)	Volcanic Tuff (fragments <2mm) Volcanic Breccia (fragments >2mm)			

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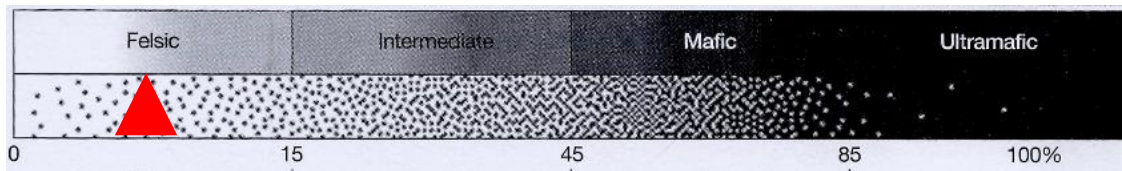


Igneous Rock Sample “A”



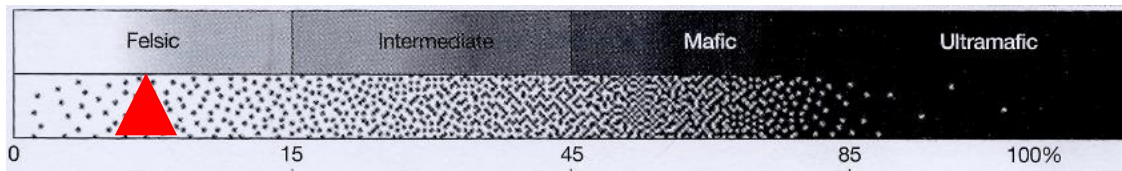


Igneous Rock Sample “B”



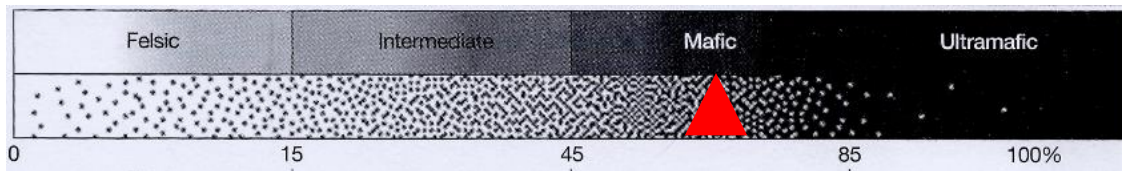


Igneous Rock Sample “C”



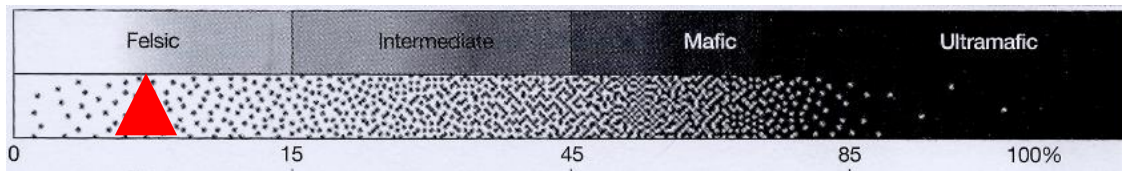


Igneous Rock Sample “D”





Igneous Rock Sample “E”





Igneous Rock Sample “F”

